

NOURHEN KHELIFI

Engineering Student in Embedded Systems & IoT

Seeking: 6-month engineering internship (final-year project), starting January 2027

+216 28 500 975 · khelifi.nourhen@gmail.com · Ariana, Tunisia
linkedin.com/in/khelifi-nourhen · github.com/khelifinourhen

PROFILE

Engineering student in Embedded Systems and Connected Objects (ISI), in the 2nd and final year of the engineering cycle. I design end-to-end systems: real-time firmware (STM32, FreeRTOS), sensor acquisition, data transport (MQTT), monitoring and mobile applications, with an AI dimension (computer vision, anomaly detection). I am looking for a 6-month engineering internship to contribute to embedded and connected products.

EDUCATION

Engineering Degree in Embedded Systems and Connected Objects 2024–present
Institut Supérieur d'Informatique (ISI), Tunisia

Bachelor's Degree in Embedded Systems 2021–2024
Institut Supérieur d'Informatique (ISI), Tunisia

Baccalaureate in Technological Sciences 2021
Lycée Sidi Hassine, Tunisia

PROFESSIONAL EXPERIENCE

Embedded Systems & IoT Engineering Intern, *YaneCode Digital* 07/2026–08/2026
- Designed a real-time monitoring chain for aeronautical parameters: acquisition (ESP32, C/C++), MQTT transport, Node-RED processing, InfluxDB storage, Grafana visualization.
- Developed a machine-learning anomaly detection module (Python).

AI Developer Intern – Computer Vision, *Sagemcom Tunisia* 07/2025–09/2025
- Developed an industrial quality-control system based on YOLOv8 and OpenCV to detect defects on GPON products.
- Built an image-processing pipeline and deployed the model on industrial data.

Junior Flutter Developer Intern, *Datadoit* 02/2024–06/2025
- Developed a smart digital-signage mobile application (Flutter/Dart, Firebase) with content management, in a collaborative workflow (Git/GitHub).

PROJECTS

Bike Tracker (in progress) – low-power embedded GPS tracker 2026
STM32L4 (B-L475E-IOT01A2), FreeRTOS (tasks, queues, semaphores), GPS NEO-6M (UART/NMEA), QSPI flash, Bluetooth

Smart Posture Monitoring System – posture monitoring with an inertial sensor and AI 2026
STM32, MPU6050 (I2C), ESP32, Node.js, MongoDB, Flutter

Connected Automatic Food Dispenser 2026
STM32L4 (bring-up, interrupts, sensors/actuators), Node.js, MongoDB, Flutter, AI

Smart Waste Bin – computer vision and Machine Learning 2023
Raspberry Pi, Machine Learning, Proteus, SolidWorks

Line-Following Robot 2022
Arduino, IR sensors, Proteus, TinkerCAD

SKILLS

Embedded systems: STM32 (FreeRTOS, interrupts, peripherals), ESP32, Raspberry Pi 4, Arduino, embedded Linux, VHDL (FPGA)
Protocols & networking: MQTT, CoAP, CAN, I2C, UART, RS232, QSPI, REST APIs
Languages: C, C++, Python, Java, Assembly, Dart, VHDL, SQL
AI & data: Machine Learning, Deep Learning (CNN, ANN), YOLOv8, OpenCV, anomaly detection
Monitoring & integration: Node-RED, InfluxDB, Grafana, Node.js, MongoDB, Flutter
Tools: Linux, Git/GitHub, Proteus, SolidWorks, Figma

CERTIFICATIONS

- CDOSS – AI and Machine Learning Fundamentals (2025) · CDOSS – Python for Data Science (2025)
- Great Learning – Introduction to Machine Learning (2024) · Microchip University – Syntax and Structure of C (2024)

LEADERSHIP & ACHIEVEMENTS

- **Codex Junior Enterprise**: President 2025–2026
- **Tunisian Confederation of Junior Enterprises**: IS Project Officer 2024–2025
- **GDSC ISI**: 2nd place (public) and 8th place (local), ML hackathon "Hack Una Matata" 2024
- **Enactus ISI**: Head of Project Development and Research Unit 2022–2023
- **Freeways ISI**: 1st place, "Freeways on Wheels 1.0" competition 2022

LANGUAGES

Arabic (native) · French (B2) · English (B2) · Spanish (A1) · German (basic)